

ECHS FRAUD ANALYTICS

Identity Fraud & Duplicate Claim Detection Analysis

COMPREHENSIVE REPORT — FY 2021-2026 EDITION (LAST 5 YEARS)

CLASSIFICATION	PERIOD	RECORDS SCANNED	CHECKS RUN	CASES FLAGGED
RESTRICTED	FY 2021- 26	26M+	6 Checks	997,624

IIT KANPUR — Data Analytics & Fraud Intelligence Division

5 June 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

EXECUTIVE SUMMARY

This report presents findings from identity and claims fraud analytics conducted on the ECHS database covering the five-year period FY 2021–2026. Six independent fraud checks were applied across **26+ million claim records** to identify duplicate identifiers, simultaneous admissions, bill resubmission, mobile fraud rings, Aadhaar (UID) duplication, and statistical over-utilisation outliers.

TOTAL CASES FLAGGED	TOTAL EXPOSURE	CHECKS W/ FINDINGS	ANALYSIS PERIOD
997,624	₹30729.39 Cr	6	5 Yrs
across 6 fraud checks	estimated financial risk	of 6 checks	FY 2021–2026

SIX FRAUD CHECKS — SUMMARY

#	Check	What It Detects	Cases Flagged	Risk
1	Duplicate Card IDs	Single ECHS card linked to 3+ different service numbers (ex-serviceman identifiers) — physical identity fraud	1,220	CRITICAL
2	Simultaneous Admissions	Same beneficiary admitted to 2 different hospitals within 7 days — physical impossibility indicating ghost billing	10	CRITICAL
3	Duplicate Bill Numbers	Identical bill/invoice number resubmitted for multiple claims — direct double-billing	250,090	HIGH
4	Mobile Number Rings	Single mobile number linked to 5+ unrelated ECHS cards — coordinated fraud agent network	177,291 real + 483 dummy	HIGH
5	UID (Aadhaar) Duplication	Same 12-digit biometric Aadhaar UID registered under multiple service numbers — synthetic identity creation	82,930	CRITICAL
6	High Frequency Claims	Beneficiaries above statistical outlier threshold (13+ claims in 5 yrs, Q3+1.5×IQR) — over-utilisation fraud	486,083	HIGH

IMMEDIATE RECOMMENDED ACTIONS

- Freeze all ECHS cards sharing a duplicate card number** (1,220 cases) — block further claims until physical verification confirms the legitimate cardholder.
- Initiate field audit of all simultaneous admission pairs** (10 cases) — cross-reference hospital admission registers against ECHS claim dates to identify ghost patients.
- Reject all duplicate bill number resubmissions** (250,090 cases) — implement system-level bill number uniqueness enforcement to prevent re-entry.
- Investigate mobile number rings** (177,291 real rings) — a single mobile coordinating 5+ cards indicates a fraud agent; de-register the mobile from non-family cards.
- Audit all UID duplication cases** (82,930 UIDs) — each Aadhaar must map to exactly one identity; escalate duplicates to UIDAI for verification and FIR filing.
- Pre-auth review for high-frequency claimants** (486,083 beneficiaries above 13 claims) — require clinical justification for further admissions.

CHECK 1

Duplicate ECHS Card Analysis
(3+ Service Numbers per Card)

DUPLICATE CARD IDS — IDENTITY FRAUD

Description: Each ECHS card is issued to one individual only. When a single card number appears linked to 3 or more distinct service numbers (ex-servicemen), it indicates the card has been cloned, stolen, or fraudulently registered under multiple identities. The card number is the primary access credential for healthcare entitlement — duplication = direct identity fraud.

1,220 cards flagged with 3+ service numbers. Top 15 by total exposure shown.

Table 1.1 — Top Duplicate Card Cases (Top 15 by Exposure)

Card Number	Svc #s	Beneficiary Names	Claims	Hospitals	Total Exposure
Card not hande	555	ABHAY RANJAN KUMAR, AJAY KUMAR SING...	1044	98	₹1.23 Cr
000002266290	3	ABHAY SINGH	22	NARAYANA HRUDAYALAYA LTD - JAIPUR, FORTIS HOSPITAL LTD - JAI...	₹50.12 L
DL200000483646	3	HEM CHANDER	49	SURBHI HOSPITAL PVT LTD-Noida, METRO HOSPITAL & HEART INSTIT...	₹47.22 L
000003451950	3	KAILASH CHANDRA RAJPUT	21	PARK HOSPITAL(UNIT OF KAILASH SUPERSPECIALITY HOSP	₹46.89 L
BR000005459074	3	BALBIR SINGH VARMA, BALBIR SINGH VE...	22	10	₹41.20 L
000006137914	3	SUBA SINGH	75	FORTIS HOSPITAL LTD - JAIPUR	₹38.65 L
000000753182	3	JAIPAL SINGH	53	OXYGEN HOSPITAL, M R MEMORIAL NAVJEEVAN HOSPITAL - ROHTAK, N...	₹37.69 L
HI000002244028	3	SATYAPAL	5	PARK HOSPITAL (A UNIT OF PARK MEDICENTRES & INSTITUTIONS PVT...	₹33.85 L
DL200000337301	3	JAS PAL SINGH RAWAT	39	METRO HOSPITAL & HEART INSTITUTE - NOIDA, Noida	₹30.68 L
JA000003018268	3	RAN TEJ	23	8	₹27.39 L
000005207621	3	LALIT	71	YATHARTH WELLNESS HOSPITAL & TRAUMA CENTRE, YATHARTH WELLNES...	₹25.83 L
000006270199	3	SURINDER SINGH	311	HEALING TOUCH SUPER SPECIALITY HOSPITAL, ROTARY AMBALA CANCE...	₹25.50 L
000002457218	3	SATYAVEER SINGH	15	5	₹25.29 L
HI000000356685	3	BALBIR SINGH	14	RS GAUR GLOBAL MULTISPECIALITY HOSPITAL PVT LTD, PARK HOSPIT...	₹24.50 L
0000816002	3	KIRAT SINGH	7	UPADHYAY HOSPITAL, PURUSHOTTAMDAS SAVITRIDEVI CANCER CARE & ...	₹24.27 L

Key Findings

- Svc #s = Unique ex-serviceman military identifiers sharing one card. Legitimate sharing covers only family members under one service number — never multiple service numbers. Any card with 3+ service numbers is a confirmed duplication.
- Total exposure represents the cumulative amount claimed using this duplicated card. Full dataset (1,220 records) available in CSV for bulk remediation.

CHECK 2

Cross-Hospital Admission Overlap
(Gap ≤ 7 Days, FY 2021–26)

SIMULTANEOUS ADMISSIONS — PHYSICAL IMPOSSIBILITY

Description: A beneficiary cannot be physically admitted to two different hospitals at the same time or within an impossibly short gap. This check identifies claim pairs from the same service number where admissions at different hospitals overlap or are separated by ≤7 days — hospital hopping to claim two separate inpatient episodes for what was a single medical event, or ghost billing where the second admission never occurred.

10 admission pairs detected with gap ≤ 7 days. Top 15 by exposure shown.

Table 2.1 — Simultaneous/Near-Simultaneous Admissions (Top 15 by Exposure)

Beneficiary	Svc #	Hospital 1	Hospital 2	Adm 1	Adm 2	Gap (Days)	Total Exposure
RAMESH NAGPAL	IC22127H	Lodhi Road Delhi 2, Delhi	NORTH EAST HEALTHCARE PVT LT Delhi, Delhi	2025-03-22	2025-06-15	0	₹1.03 Cr
ROHTASH KUMAR	13944311K	FORTIS HOSPITALS LTD - FARID Delhi, Delhi	METRO HEART INSTITUTE - FARI Delhi, Delhi	2023-07-03	2023-07-05	2	₹73.32 L
ARUN KUMAR SHOREY	IC36348A	MANIPAL HOSPITALS PRIVATE LI Delhi, Delhi	Gurgaon (Dhundahera) Delhi, Delhi	2023-10-15	2023-11-12	6	₹68.75 L
NARESH KUMAR BUDHWAR	675280B	Rohtak Hisar, Haryana	Secunderabad 2nd(Bowenpally) Secunderabad, Andhra	2023-06-29	2023-08-16	4	₹62.64 L
OM SINGH	2977308W	RDM HOSPITAL Jaipur, Rajasthan	HEALTHCARE GLOBAL ENTERPRISE Jaipur, Rajasthan	2025-01-01	2025-01-06	5	₹58.11 L
JAGAN NATH	2861394N	MEDANTA THE MEDICITY - GURGA Delhi, Delhi	JAYPEE HEALTHCARE LIMITED - Delhi 2, Delhi	2022-05-05	2022-09-02	0	₹56.43 L
RAM PRAKASH	10315725N	Bulandshahr Bareilly, Uttar Prad	KAILASH HOSPITAL & HEART INS Delhi 2, Delhi	2024-06-07	2024-07-07	2	₹54.31 L
DAMBAR SINGH	4155896A	FORTIS HOSPITAL - NOIDA Delhi 2, Delhi	MEDANTA THE MEDICITY - GURGA Delhi, Delhi	2025-07-15	2025-08-13	1	₹54.00 L
BRAJ NARAYAN SINGH	2999300K	NIRMAL HOSPITAL Jabalpur, Madhya Pra	MEDANTA THE MEDICITY - GURGA Delhi, Delhi	2023-03-13	2023-03-18	1	₹52.85 L
BRAJ NARAYAN SINGH	2999300K	MEDANTA THE MEDICITY - GURGA Delhi, Delhi	NIRMAL HOSPITAL Jabalpur, Madhya Pra	2023-03-18	2023-06-07	3	₹52.80 L

Key Findings

- Gap/Overlap Days: Negative values indicate a true simultaneous overlap — the beneficiary was admitted to Hospital 2 before being discharged from Hospital 1. Zero = same-day discharge and admission. Both are physical impossibilities for genuine inpatient care.
- Cases where Hospital 2 name shows as a short code (e.g. "Rohtak", "Lodhi Road") indicate ECHS polyclinics — referral-chain fraud where a polyclinic and a private hospital both file claims for the same episode of care.

CHECK 3

Bill/Invoice Number Resubmission
(Excluding NA / Blank)

DUPLICATE BILL NUMBERS — RESUBMISSION FRAUD

Description: Each hospital bill or invoice carries a unique number. If the exact same bill number is submitted under multiple ECHS claim IDs, it means either: (a) the same bill was physically resubmitted for payment twice, or (b) the bill number field was copied from another claim — both are direct billing fraud. All blank, "NA", and null bill numbers are excluded from this analysis.

250,090 duplicate bill numbers detected. Top 15 by duplicate count shown.

Table 3.1 — Most Resubmitted Bill Numbers (Top 15)

Bill Number	Dup Count	Claims	Hospitals	Beneficiary/Svc #	Total Exposure
202436702	NULL	1	LU000001767189	31825498, 31825499, 31825500 +75 more	₹78
305101	2	2	Ara Bhojapur	SURENDRA SINGH	₹88.57 L
221338	2	2	Bangalore (Deactivated)	S ANANDA RAO	₹57.18 L
IPCS24000819	3	3	Vijayawada	SUNKAVALI SREE RAMMOHAN	₹57.00 L
1003047823	2	2	Quilon (Kollam)	MANOJ V	₹54.70 L
MILL - ICS - 63388	34	34	LUCKNOW	AWDH NARESH MISRA	₹51.12 L
GHIPBL/ 25091060	2	2	Chandigarh	JOGA SINGH	₹50.61 L
4085	3	3	Perinthelmann	ZACHARIAS P C	₹49.80 L
BILL ATTACHED	178	94	Delhi Cantt (BHDC), Gurgaon (Dhundaheera), Gurgaon (Soha...	RAJENDER KUMAR, SHRI KRISHAN GARG, JAI BHAGWAN +37...	₹46.69 L
9197	2	2	Palampur (RC Yol)	OM PRAKASH	₹46.08 L
MHIC177101	3	3	Chandimandir	MC DOGRA, OM PRAKASH LAMBA	₹44.23 L
IPCS23001710	2	2	Secunderabad Main	TARKESHWAR SINGH	₹42.00 L
3002/Bill/ECHS	26	26	Dehradun	BHIM SINGH NEGI, POORAN SINGH ASWAL, KUNWAR SINGH ...	₹41.61 L
SUMMARY ATTACHED	29	29	Kanpur Nagar	HARISH CHANDRA VARMA	₹39.81 L
PTB/800495	2	2	Pune	TUKARAM BABAN GAWARE	₹39.33 L

Key Findings

- Dup Count** = number of distinct ECHS claims carrying the same bill string. A count of 2 = one resubmission; counts of 5+ indicate systematic reuse of the same document across many claims — a template-based billing fraud operation.
- Bill number duplicates appearing across multiple hospitals (Hospitals column > 1) are particularly severe: the same physical bill cannot have been issued by two separate facilities, confirming fabrication of the invoice document itself.

CHECK 4

Mobile Coordination Network
(5+ Cards per Mobile, FY 2021–26)

MOBILE NUMBER RINGS — COORDINATED FRAUD AGENT NETWORK

Description: An ECHS mobile number is used for claim communication and OTP verification. Legitimate sharing within a family (one ex-serviceman + dependents) produces at most 3–4 cards on one number. When a single mobile is linked to 5+ cards belonging to different service numbers, it indicates a **fraud agent** operating as the contact point for multiple unrelated identities — filing claims on behalf of ghost beneficiaries. **483 dummy numbers** (all-zero, repeating digits) were excluded and reported separately below.

177,291 real mobile rings detected. Top 15 by total exposure shown.

Table 4.1 — Top Mobile Number Rings (Real Numbers, Top 15 by Exposure)

Mobile Number	Cards	Svc Numbers	Claims	Hospitals	Total Exposure
9599752407	6	2	204	18	₹2.99 Cr
8527425823	10	1	95	GUPTA MULTISPECIALITY HOSPITAL, SRI BALAJI ACTION MEDICAL IN...	₹2.84 Cr
9045001845	5	2	251	11	₹2.81 Cr
9745453798	6	1	182	KERALA INSTITUTE OF MEDICAL SCIENCES, Chaithanya Eye Hospita...	₹2.70 Cr
9910135754	5	2	138	12	₹2.57 Cr
7027355088	5	1	214	ARTEMIS MEDICARE SERVICES LIMITED - GURGAON, PUSHPANJALI HOS...	₹2.57 Cr
9557421990	5	1	104	7	₹2.53 Cr
9812058813	5	4	68	ROTARY AMBALA CANCER & GENERAL HOSPITAL, INDUS INTERNATIONAL...	₹2.52 Cr
9634998085	6	1	222	7	₹2.45 Cr
9915959886	6	1	111	MANIPAL HOSPITAL PVT LTD, AMAR HOSPITAL - PATIALA, SHALBY HO...	₹2.43 Cr
9779829929	6	1	83	8	₹2.42 Cr
9466230981	5	2	162	14	₹2.36 Cr
8851313282	15	1	164	14	₹2.34 Cr
9540367061	5	1	163	8	₹2.34 Cr
9467108388	8	3	122	12	₹2.34 Cr

Table 4.2 — Dummy/Invalid Mobile Numbers (Excluded from Main Analysis)

Mobile Number	Cards	Claims	Total Exposure
0000000000	2649	4445	₹9.97 Cr
9999999999	642	1303	₹1.58 Cr
9999156421	5	49	₹1.17 Cr
9999265464	5	544	₹70.01 L

Mobile Number	Cards	Claims	Total Exposure
9999150933	5	41	₹65.91 L
9999971626	10	358	₹60.05 L
1234567890	186	840	₹58.01 L
9999762583	8	117	₹53.25 L
9999390379	7	132	₹52.68 L
9999680725	5	11	₹51.57 L

Key Findings

- Cards** = distinct ECHS health cards registered under this single mobile. Fraud agents maintain one phone number as their operational contact across all ghost identities they manage — allowing them to receive OTPs, discharge summaries, and claim status alerts centrally.
- Dummy numbers** (Table 4.2) represent cases where the mobile field was filled with placeholder data (000000, 111111, etc.) — indicating the beneficiary has no real mobile on file, making contact and verification impossible.

CHECK 5

Aadhaar UID Cross-Linkage Analysis
(Dummy UIDs Excluded)

UID DUPLICATION — SYNTHETIC IDENTITY FRAUD

Description: The 12-digit Aadhaar UID is a biometric-linked national identifier — exactly one UID per individual, guaranteed by UIDAI. If the same UID appears registered under multiple ECHS service numbers, it means: (a) one person's identity is being used by another (**identity theft**), or (b) a single UID has been bulk-registered under fabricated service numbers to generate fictitious beneficiaries (**synthetic identity fraud**). Known dummy UIDs (all-zero, all-nine, sequential) are excluded. All 12-digit UIDs masked as XXXX****XXXX in the report.

82,930 UIDs found linked to multiple service numbers. Top 15 by exposure shown.

Table 5.1 — UID Duplication Cases (Top 15 by Service Number Count)

UID (masked)	Svc #s	Claims	Hospitals	Locations	Total Exposure
7561****7493	60	66	Delhi Cantt (BHDC), KALRA HOSPITAL SRCNC PVT LTD - KIRTI NAG...	Delhi-Delhi	₹1.37 Cr
9456****4987	98	111	KALRA HOSPITAL SRCNC PVT LTD - KIRTI NAGAR	Delhi-Delhi	₹1.29 Cr
7379****6557	2	49	Bhubaneswar (RC Ranchi), Bhubaneswar (RC Bhubneshwar)	Bhubaneswar-Rajasthan Ranchi-Odis	₹1.19 Cr
9429****0803	204	442	MITTAL HEALTHCARE, TATA MEMORIAL CENTRE (SANGRUR), HOMI BHAB...	Ambala-Punjab Chandimandir-Chandi	₹1.16 Cr
8380****7991	44	59	6	Delhi-Delhi Delhi 2-Delhi Hisar	₹1.05 Cr
6757****9581	9	37	MAX SUPER SPECIALITY HOSPITAL - SHALIMAR BAGH, KALRA HOSPITA...	Delhi-Delhi	₹1.00 Cr
2078****4437	3	81	Noida	Delhi 2-Delhi	₹95.51 L
8697****3543	2	13	METRO HOSPITAL & CANCER INSTITUTE - PREET VIHAR	Delhi 2-Delhi	₹85.52 L
2190****3566	264	517	Bengdubi	Kolkata-West Bengal	₹85.35 L
5386****5680	2	40	18	Delhi-Delhi Delhi 2-Delhi	₹82.21 L
7852****1077	2	8	5	Allahabad-Uttar Pradesh Delhi 2-D	₹80.85 L
9289****1453	26	34	SHAKUNTLA NURSING HOME & HOSPITAL, KALRA HOSPITAL SRCNC PVT ...	Delhi-Delhi	₹78.84 L
2360****8882	42	54	HCG CANCER CENTRE	Ahmedabad-Gujarat	₹78.76 L
9840****5751	31	40	ORTHO PLUS HOSPITAL, NKS HOSPITAL (A UNIT OF BRAM HEALTH CAR...	Delhi-Delhi	₹77.34 L
8185****2562	32	39	PARK HOSPITAL - FARIDABAD, NKS HOSPITAL (A UNIT OF BRAM HEAL...	Delhi-Delhi	₹76.99 L

Key Findings

Svc #s = count of distinct military service numbers sharing this UID. Each UID should map to exactly **one** individual. Even two service numbers sharing one UID is anomalous — it means one person's biometric is funding two ECHS accounts.

UID duplication cases involving 3+ service numbers and multiple geographic locations indicate a **large-scale synthetic identity ring** — a single Aadhaar is the seed identity for an entire cluster of fabricated ECHS accounts. These must be escalated to UIDAI and ECHS Directorate.

CHECK 6

Statistical Outlier Detection
(Q3 + 1.5×IQR Threshold: 13 claims)

HIGH FREQUENCY CLAIMS — OVER-UTILISATION FRAUD

Description: The threshold for this check was not fixed arbitrarily — it was **calculated dynamically** from the actual distribution of all 4.69 million beneficiaries over 5 years. Statistical analysis produced: Median = 2 claims, Q3 = 6 claims, IQR = 5, giving an outlier threshold of **13+ claims** (Q3 + 1.5×IQR). Any beneficiary above this boundary is a statistical outlier — 1 in 1,000+ probability of occurring by legitimate clinical need alone. This indicates planned over-utilisation, ghost visit billing, or collusion with a hospital.

486,083 beneficiaries exceeded the 13-claim threshold. Top 15 by total exposure shown.

Table 6.1 — High Frequency Claimants Above Threshold (Top 15 by Exposure)

Service #	Beneficiary	# Claims	Avg Claim	Hospitals	Total Exposure
6285365Y	BHUPAL SINGH	143	₹5.20 L	FORTIS HOSPITAL - SHALIMAR BAGH	₹7.13 Cr
15766301H	AJAYA KUMAR NAYAK	46	₹13.25 L	KALINGA INSTITUTE OF MEDICAL SCIENCES, KALINGA HOSPITAL LTD	₹5.96 Cr
4003982F	PIARE LAL	49	₹11.45 L	SHALBY HOSPITAL - MOHALI, Fortis Hospital - Mohali	₹5.50 Cr
2986165A	SHRAWAN GUJAR	104	₹4.20 L	5	₹4.32 Cr
JC240738K	JOGINDER SINGH	221	₹1.98 L	8	₹4.27 Cr
4447911\$	MALKIT SINGH	103	₹3.51 L	FORTIS HOSPITAL - LUDHIANA, HOMI BHABHA CANCER HOSPITAL & RE...	₹3.58 Cr
792369G	RANJITHKUMAR M B	128	₹2.57 L	MVR CANCER CENTER RESEARCH INSTITUTE, AMALA CANCER HOSPITAL ...	₹3.24 Cr
725915N	SATYAVIR SINGH	89	₹3.82 L	7	₹3.21 Cr
JC325283W	NANDA BALLABH TIWARI	70	₹4.64 L	METRO HOSPITAL & HEART INSTITUTE - NOIDA, METRO HOSPITAL & C...	₹3.20 Cr
4177774W	PUSHKAR SINGH JETHI	162	₹2.00 L	MEDANTA THE MEDICITY - GURGAON, UJALA CYGNUS CENTRAL HOSPITA...	₹3.20 Cr
JC538107F	KHIM SINGH	76	₹4.30 L	5	₹3.18 Cr
JC135912M	KASHMIR SINGH	66	₹4.68 L	CAPITOL HOSPITAL, FORTIS ESCORTS HOSPITAL (A UNIT OF FHL & I...	₹3.09 Cr
14566017M	KRISHAN PAL YADAV	100	₹2.97 L	MAX SMART SUPER SPECIALITY HOSPITAL - SAKET, MAX SMART SUPER...	₹2.94 Cr
3378042A	PARAS RAM	65	₹4.66 L	NARAYANA VAISHNO DEVI SPECIALITY HOSPITALS PRIVATE, RAKSHA K...	₹2.94 Cr
4457835A	RAGHBIR SINGH	112	₹2.59 L	AMAR HOSPITAL - PATIALA, SIMRITA NURSING HOME, NEELAM HEALTH...	₹2.87 Cr

Key Findings

The statistical baseline (median = 2 claims over 5 years) reflects the typical ECHS beneficiary pattern. A beneficiary with 13+ claims is 6× above the median. Beneficiaries with 30+ claims over 5 years (6 per year) warrant immediate clinical audit — at that frequency, inpatient claims can no longer be explained by age or chronic illness alone.

High-frequency claimants concentrated at a **single hospital** (Hospitals = 1) are the most suspicious: genuine chronic illness patients typically see multiple specialists across different facilities. One hospital + high frequency = hospital-patient collusion for ghost billing.

CONSOLIDATED RISK REGISTER

All findings are based on structured database analysis and must be corroborated with physical audit records before enforcement action is taken.

Check	Fraud Signal	Cases Flagged	Exposure	Risk
1	Duplicate Card IDs (3+ Svc Numbers)	1,220	₹30.89 Cr	CRITICAL
2	Simultaneous Admissions (Gap ≤ 7 days)	10	₹6.36 Cr	CRITICAL
3	Duplicate Bill Number Resubmission	250,090	₹948.78 Cr	HIGH
4	Mobile Number Rings (5+ cards)	177,291 real + 483 dummy	₹11063.98 Cr	HIGH
5	UID (Aadhaar) Duplication	82,930	₹1818.60 Cr	CRITICAL
6	High Frequency Claims (≥13 claims)	486,083	₹16860.78 Cr	HIGH
TOTAL		997,624	₹30729.39 Cr	—

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